Fiber optic communications

Phys 382

Spring Semester

2025-2026

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CamScanner

Physical Properties of light

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Geometrical optics theory of Light

- Geometrical optics is used to explain reflection and refraction
- In geometrical optics, light propagates as rays in straight lines that carries energy.
- At an interface reflection and refraction takes place id the angle of incidence is less than the critical angle

Snell's law

At refraction $n_1 \sin \phi_1 = n_2 \sin \phi_2$

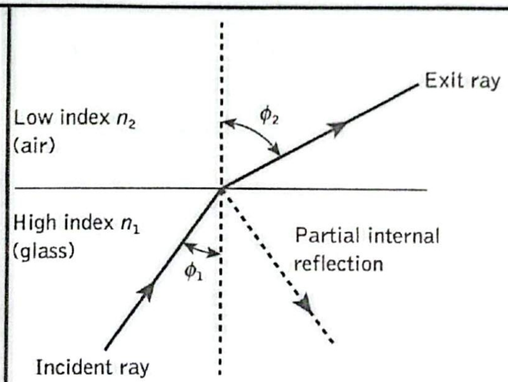
At reflection $\phi_1 = \phi_2$

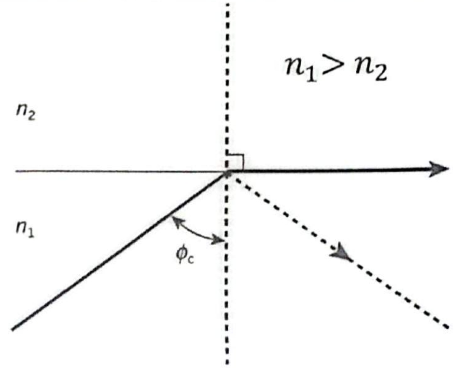
Critical angle: Light is incident from a medium with higher refractive index to a medium with lower index of refraction

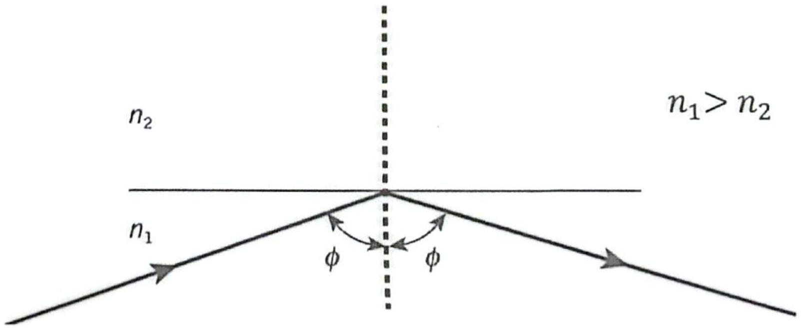
$$\sin \phi_c = \frac{n_2}{n_1}$$

Conditions for total internal reflection

1. Light is incident from higher index medium into lower index medium.
2. Incident angle is equal or larger than the critical angle.







Basic concepts of Electromagnetic theory of Light

- Electromagnetic theory considers light as electromagnetic waves.
- Electromagnetic waves consists of electric field and magnetic fields. Both are perpendicular to each other. Additionally, each is perpendicular to the direction of propagation.
- Light is transvers electromagnetic waves.
- Speed of light in vacuum is $c = 3x10^8\ m/s$
- Speed of light is a medium for monochromatic light is described by phase velocity.
- Phase velocity is the velocity of single electromagnetic wave (single wavelength)
- Most light consists of group of frequencies.
- Propagation of light in a material should be described by group velocity.

Group velocity is the velocity of a group of waves with different frequencies.

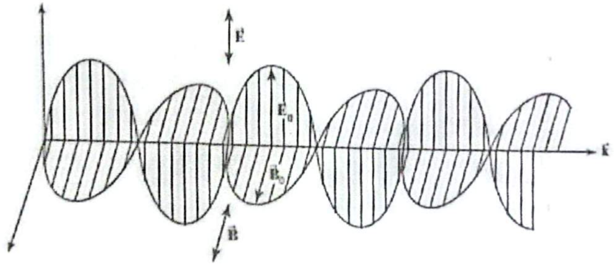
Wavelength λ: distance a constant phase point on the wave travels in one period.

Phase velocity $v_p = \frac{\lambda}{T}$

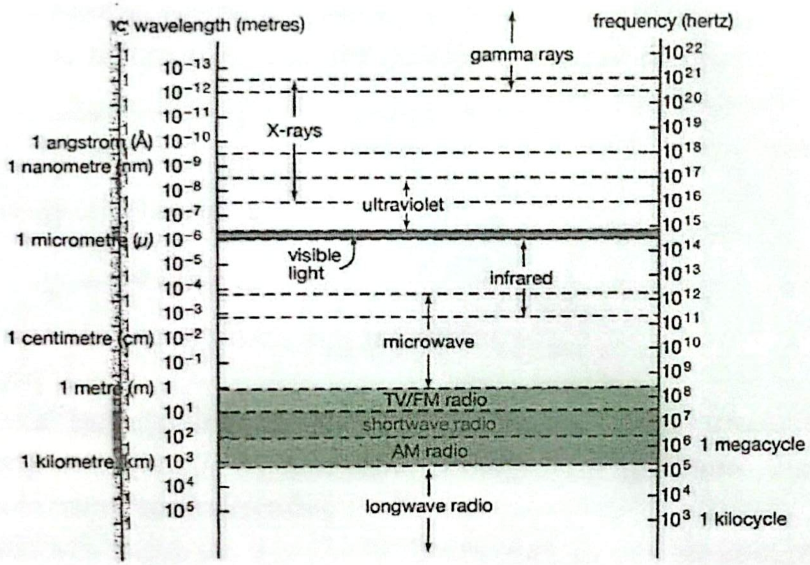
Refractive index $n = \frac{c}{v_p}$

Wavelength in a medium is $\lambda = \frac{\lambda_o}{n}$

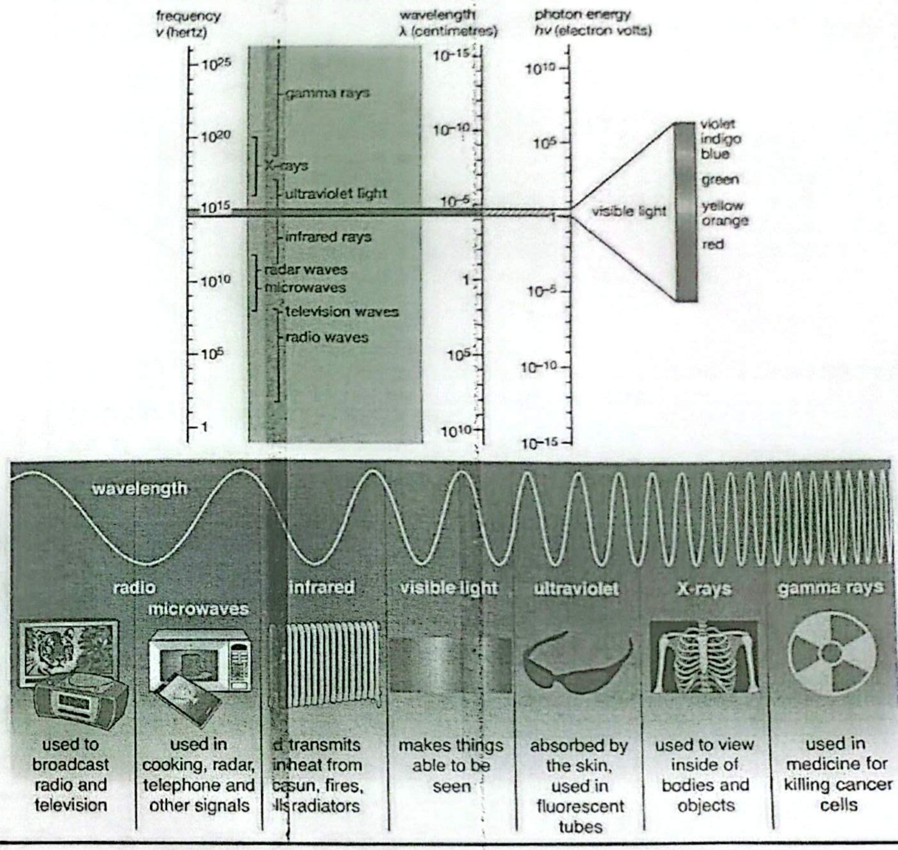
Frequency $\nu = \frac{v_p}{\lambda}$



Electromagnetic Spectrum



https://www.britannica.com/science/electromagnetic-spectrum



https://www.britannica.com/science/electromagnetic-spectrum

Electromagnetic Waves

- A plane electromagnetic wave traveling in some arbitrary direction consists of electric field $\vec{\mathbf{E}}$ and magnetic field $\vec{\mathbf{B}}$. The electric and magnetic fields are always perpendicular to one another and to the direction of propagation.
- Each component of the wave travels with the same propagation vector and frequency and thus with the same wavelength and speed.
- At any specified time t and location r

$$\vec{\mathbf{E}} = \vec{\mathbf{E}}_0 \sin(\vec{\mathbf{k}}\cdot\vec{\mathbf{r}} - \omega t) \qquad E = cB$$

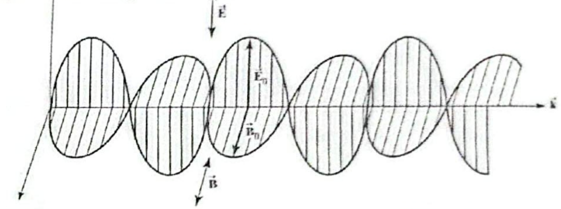
$$\vec{\mathbf{B}} = \vec{\mathbf{B}}_0 \sin(\vec{\mathbf{k}}\cdot\vec{\mathbf{r}} - \omega t) \qquad E_0 = cB_0$$

$$c = \frac{1}{\sqrt{\varepsilon_0\mu_0}}, \quad k = \frac{2\pi}{\lambda}, \quad \nu = \frac{1}{T}, \quad v = \nu\lambda, \quad \omega = 2\pi\nu$$

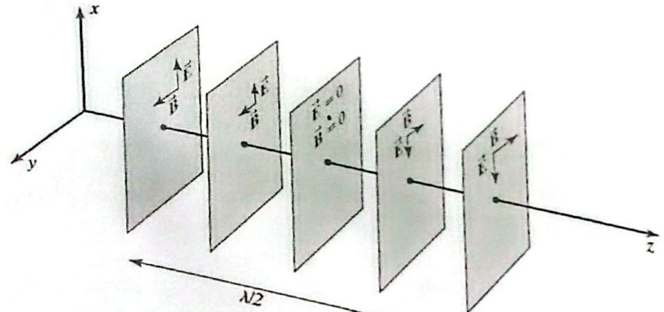
- $\vec{\mathbf{r}}$ is the position vector, E_0: amplitude of the electric field, B_0 is the amplitude of the magnetic field, ω is the angular frequency, c: speed of light in vacuum (3×10^{8} m/s), ε_0 and μ_0: permittivity and permeability of free space, respectively. ν: frequency, λ wavelength, T: periodic time.

$$\varepsilon_0 = 8.8542 \times 10^{-12}(\mathrm{C\cdot s})^2/\mathrm{kg\cdot m^3} \text{ and } \mu_0 = 4\pi \times 10^{-7}\,\mathrm{kg\cdot m/(A\cdot s)^2}.$$

Frank J. Pedrotti, Leno M, Leno S. Pedrotti, Introduction to Optics, Benjamin Cummings.Sons. 2006.



A plane wave propagating along the positive *z*-direction with the electric field varying along the *x*-direction and the magnetic field varying along the *y*-direction.



Frank J. Pedrotti, Leno M, Leno S. Pedrotti, Introduction to Optics, Benjamin Cummings.Sons. 2006.

Wavefronts for a plane electromagnetic wave.

Electromagnetic waves

- The energy density u_E, is associated with the electric field in free space is

$$u_E = \frac{1}{2}\varepsilon_0 E^2$$

- The energy density u_B associated with the magnetic field in free space is

$$u_B = \frac{1}{2}\frac{1}{\mu_0}B^2$$

$$u_B = \frac{1}{2}\frac{1}{\mu_0}B^2 = \frac{1}{2}\frac{1}{\mu_0}\left(\frac{E}{c}\right)^2 = \left(\frac{1}{2}\frac{\varepsilon_0\mu_0}{\mu_0}\right)E^2 = \frac{1}{2}\varepsilon_0 E^2 = u_E$$

- The total energy density

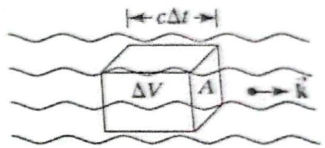
$$u = u_E + u_B = 2u_E = 2u_B$$

$$u = \varepsilon_0 E^2 = \left(\frac{1}{\mu_0}\right)B^2$$

$$u = \sqrt{u}\sqrt{u} = (\sqrt{\varepsilon_0}E)\left(\frac{B}{\sqrt{\mu_0}}\right) = \frac{\varepsilon_0}{\sqrt{\varepsilon_0\mu_0}}EB = \varepsilon_0 cEB$$

- The power is defined as the energy per unit area per unit time

$$\text{power} = \frac{\text{energy}}{\Delta t} = \frac{u\,\Delta V}{\Delta t} = \frac{u(Ac\,\Delta t)}{\Delta t} = ucA$$



Frank J. Pedrotti, Leno M, Leno S. Pedrotti, Introduction to Optics, Benjamin Cummings.Sons. 2006.

$$S = uc$$

$$S = \varepsilon_0 c^2 EB$$

- Poynting vector $\vec{\mathbf{S}}$: power per unit area in the direction of propagation.

$$\vec{\mathbf{S}} = \varepsilon_0 c^2 \vec{\mathbf{E}} \times \vec{\mathbf{B}}$$

- The irradiance E_E associated with the electric field in free space is the average value of the magnitude of the pointing vector

$$E_e = \langle|\vec{\mathbf{S}}|\rangle = \varepsilon_0 c^2 \langle E_0 B_0 \sin^2(\vec{\mathbf{k}}\cdot\vec{\mathbf{r}} \pm \omega t)\rangle$$

Electromagnetic waves

$$E_e = \frac{1}{2}\varepsilon_0 c^2 E_0 B_0$$

$$E_e = \frac{1}{2}\varepsilon_0 c E_0^2$$

$$E_e = \frac{1}{2}\left(\frac{c}{\mu_0}\right) B_0^2$$

Example 1

A beam of radius 1 mm carries a power of 6 kW. Determine its average irradiance and the amplitude of its E and B fields.

Solution

$$E_e = \frac{\text{power}}{\text{area}} = \frac{6000}{\pi(10^{-3})^2} = 1.91 \times 10^9\ \text{W/m}^2$$

$$E_0 = \left(\frac{2E_e}{\varepsilon_0 c}\right)^{1/2} = \left[\frac{2(1.91 \times 10^9)}{\varepsilon_0 c}\right]^{1/2} = 1.20 \times 10^6\ \text{V/m}$$

$$B_0 = \frac{E_0}{c} = \frac{1.20 \times 10^6}{c} = 4.00 \times 10^{-3}\ \text{T}$$

Example 2

A traveling electromagnetic wave has an electric field given by

$$y(x,t) = (0.35\ \text{m}) \sin[(3\pi/\text{m}) \times - (10\pi/\text{s})t + \pi/4)]$$

Determine the wavelength, frequency, velocity, and initial phase angle. Also find the displacement at x=10 cm and t=0

Solution

$$E = Eo \sin(kx \pm wt + \phi o)$$

$k = 3\pi/\text{m}$ and $\omega = 10\pi/\text{s}$.

$$\lambda = \frac{2\pi}{k} = \frac{2}{3}\ \text{m} \quad \text{and} \quad \nu = \frac{\omega}{2\pi} = 5\ \text{Hz}$$

$v = \lambda\nu = (2/3)5\ \text{m/s} = 3.33\ \text{m/s}$ in the positive x-direction (due to the negative sign in the phase)

The initial phase $(x = 0, t = 0)$ is $\pi/4$

$$E = (0.1\ \text{m}, 0) = (0.35\ \text{m}) \sin\left(0.3\pi + \frac{\pi}{4}\right) = +0.346\ \text{m}$$

Phase and group velocities

- Phase velocity of an electromagnetic signal is a measure of the velocity of the harmonic waves that constitute the signal.

$$v_p = \frac{\omega_p}{k_p} = \frac{\omega_1 + \omega_2}{k_1 + k_2} \cong \frac{\omega}{k} \quad \text{and} \quad v_p = c/n.$$

- *Group velocity* of the signal is the velocity at which the positions of maximal constructive interference propagate.

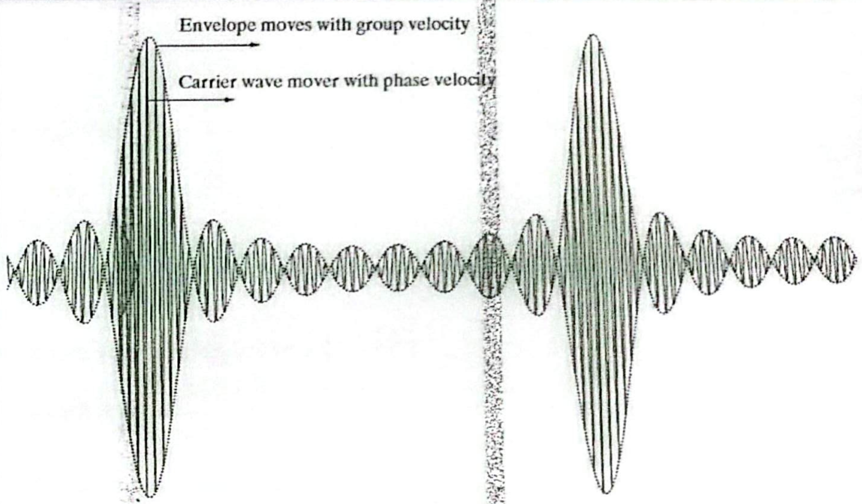
$$v_g = \frac{\omega_g}{k_g} = \frac{\omega_1 - \omega_2}{k_1 - k_2} \cong \frac{d\omega}{dk}$$

- Relationship between phase velocity and group velocity

$$v_g = v_p + k\left(\frac{dv_p}{dk}\right)$$

<u>Non Dispersive medium</u> (vacuum)

- the wavelength of light and hence the propagation consistent does not depend on frequency



<u>Dispersive medium</u>

- Refractive index is function wavelength
- Phase velocity is function of wavelength
- Group velocity is not equal to phase velocity.

$$v_g = v_p\left[1 + \frac{\lambda}{n}\left(\frac{dn}{d\lambda}\right)\right]$$

Phase and group velocities

Example 3

For wavelengths in the visible spectrum, the index of refraction of a certain type of crown glass can be approximated by the relation $n(\lambda) = 1.5255 + (4825\ \text{nm}^2)/\lambda^2$.

a. Find the index of refraction of this glass for 400 nm light, 500 nm light, and 700 nm light.

b. Plot the refractive index as a function of wavelength over the visible spectrum (400 nm-700 nm).

d. Find the phase velocity, in this glass, for a pulse with frequency components centered around 500 nm.

c. Find the group velocity, in this glass, for a pulse with frequency components centered around 500 nm.

a. The indices are

$$n_{400} = 1.5255 + (4825\ \text{nm}^2)/(400\ \text{nm})^2 = 1.5557$$

$$n_{500} = 1.5255 + (4825\ \text{nm}^2)/(500\ \text{nm})^2 = 1.5448$$

$$n_{700} = 1.5255 + (4825\ \text{nm}^2)/(700\ \text{nm})^2 = 1.5353$$

b. The phase velocity is $v_p = \dfrac{c}{n_{500}} = \dfrac{3 \times 10^8\ \text{m/s}}{1.5448} = 1.942 \times 10^8\ \text{m/s}$.

c. The group velocity for this pulse would be

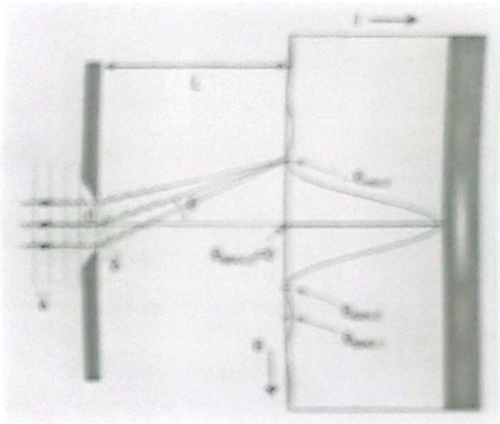
$$v_g = v_p\left[1 + \frac{\lambda}{n}\left(\frac{dn}{d\lambda}\right)\right]\Bigg|_{\lambda=500\ \text{nm}} = v_p\left[1 + \frac{\lambda}{n}\left(\frac{4825\ \text{nm}^2}{\lambda^3}\right)(-2)\right]\Bigg|_{\lambda=500\ \text{nm}}$$

$$= v_p\left[1 - \frac{9650\ \text{nm}^2}{n\lambda^2}\right]\Bigg|_{\lambda=500\ \text{nm}}$$

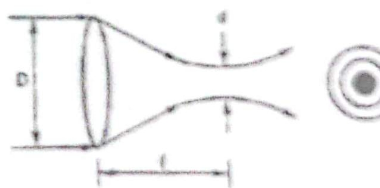
$$v_g = (1.942 \times 10^8\ \text{m/s})\left[1 - \frac{9650\ \text{nm}^2}{(1.5448)(500\ \text{nm})^2}\right] = 1.893 \times 10^8\ \text{m/s}$$

Diffraction

- Diffraction is the deviation of waves from straight-line propagation without any change in their energy due to an obstacle or through an aperture.
- In geometrical optics, a collimated beam is focused by a perfect lens into a spot rather than a point because of diffraction. The central spot size d
- $d = \frac{2.44\lambda f}{D}$
- The central spot is usually small.
- Maybe neglected in some applications (geometrical optics is good enough in these applications).
- In the case of coupling light into the core of a fiber
- Coupling efficiency is low if light is coupled into a glass of thickness that is less than the spot size.
- Diffraction is required to explain the pattern that is formed.



https://en.wikipedia.org/wiki/Diffraction



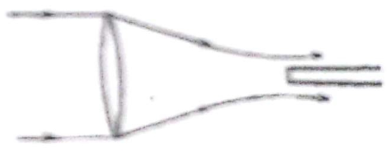
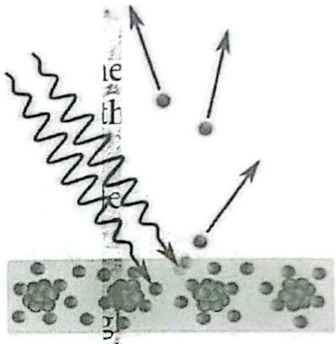


Figure 2-25 Focusing a beam onto a sma
may result in inefficient coupling.

Basic concepts of Quantum theory of light

- Light exhibits particle-like behavior when exchanging energy with matter, as in the photoelectric effects.
- Light propagates as packets of light energy called photons.
- Each photon has energy $E = h\nu$
- Einstein explained the photoelectric effect: light is a stream of light quanta whose energy is related to frequency by Planck's equation.
- Photoelectric effect is the emission of electrons from a metal surface when irradiated with light.
- Louis de Broglie: a particle with momentum p had an associated wavelength



https://en.wikipedia.org/wiki/Photoelectric_effect

$$\lambda = \frac{h}{p}$$

$p = mv$

h is planck's constant

h=6.63 * 10^{-34} J-s.

Basic concepts of Quantum theory of light

Example 4

A certain sensitive radar receiver detects an electromagnetic signal of frequency 100 MHz and power (energy/time)

a. What is the wavelength of a photon with this frequency?

b. What is the energy of a photon in this signal in Joules?

c. How many photons in this signal?

d. What is the energy in eV of a visible photon of wavelength 555 nm?

e. How many visible photons/s would correspond to a detected power of 6.63 x 10-16 J/s?

f. What is the energy Joules of an X-ray of wavelength 0.1 nm?

Where, h=6.62607015×10^{-34} J-Hz^{-1}

Energy of one photon= $h\nu$

Solution

a. $\lambda = c/\nu = \dfrac{3 \times 10^8 \text{ m/s}}{100 \times 10^6 \text{ Hz}} = 3 \text{ m}$

b. $E = h\nu = (6.63 \times 10^{-34} \text{ J}\cdot\text{s})(100 \times 10^6 \text{ Hz}) = 6.63 \times 10^{-26} \text{ J}$

$$E = 6.63 \times 10^{-26} \text{ J}\left(\frac{1 \text{ eV}}{1.6 \times 10^{-19} \text{ J}}\right) = 4.14 \times 10^{-7} \text{ eV}$$

c. The number of detected photons per second N would be

$$N = \frac{\text{Power}}{\text{Energy/photon}} = \frac{6.63 \times 10^{-16} \text{ J/s}}{6.3 \times 10^{-26} \text{ J}} = 10^{10}/\text{s}$$

d. $E_{555} = h\nu = \dfrac{hc}{\lambda} = \dfrac{(6.63 \times 10^{-34} \text{ J}\cdot\text{s})(3 \times 10^8 \text{ m/s})}{555 \times 10^{-9} \text{ m}}$

$= 3.58 \times 10^{-19} \text{ J} = 2.2 \text{ eV}$

e. $N_{555} = \dfrac{\text{Power}}{\text{Energy/photon}} = \dfrac{6.63 \times 10^{-16} \text{ J/s}}{3.58 \times 10^{-19} \text{ J}} = 1850/\text{s}.$

f. $E_{\text{X-ray}} = h\nu = \dfrac{hc}{\lambda} = \dfrac{(6.63 \times 10^{-34} \text{ J}\cdot\text{s})(3 \times 10^8 \text{ m/s})}{0.1 \times 10^{-9} \text{ m}}$

$= 1.99 \times 10^{-15} \text{ J} = 12{,}400 \text{ eV}$

Physical properties of light

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2. Introduction to Optics, by Frank J. Pedrotti, Leno M, Leno S. Pedrotti, 3rd ed. 2006, Publisher: Benjamin Cummings. Sons. 2005.

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5. https://en.wikipedia.org/wiki/Photoelectric_effect

https://www.britannica.com/science/electromagnetic-spectrum